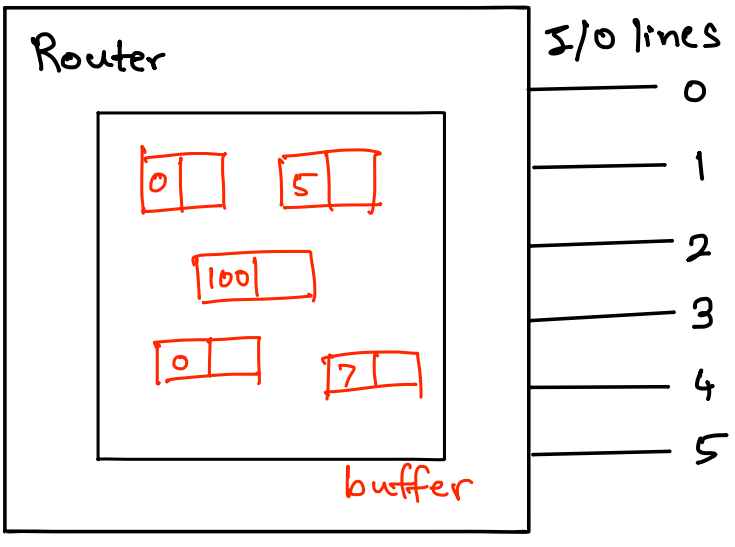
COL1101 Lecture 16 2026/08/27

Topic: Priority Queues

Announcements

Exam OS testing in LH ~~325~~ 121

Motivating Example



- Receive incoming packets in buffer.
- Send packets one-by-one.

  Most urgent first

Recap: Queues

- Enqueue
- Dequeue
  (First-in-first-out)
- isEmpty, len, etc

Want: Priority Queues

- Enqueue
- Extract Min
  (min-first-out)
- is Empty, len, etc.

Implementations?

	Enqueue	Extract Min
(Linked) List in sorted order	$\Theta(n)$	$\Theta(1)$
Unordered (Linked) List	$\Theta(1)$	$\Theta(n)$